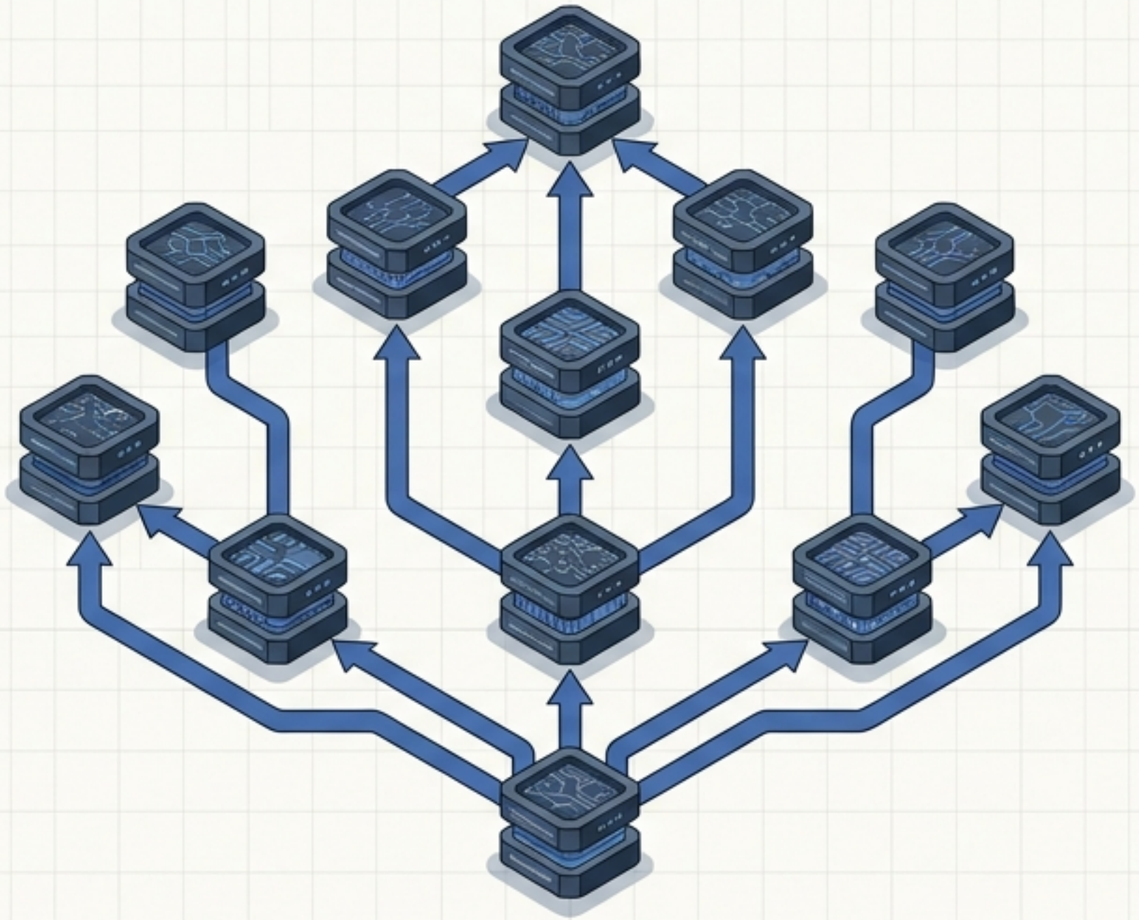


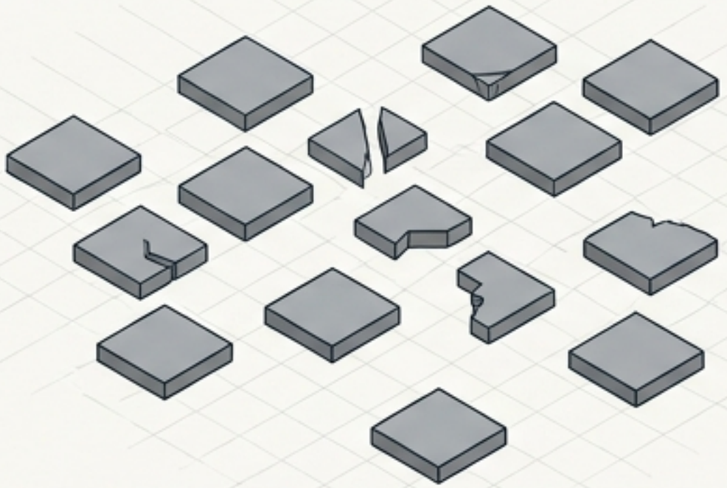
agent-team-harness

One Persona. One Memory. One Set of Lessons.

Gemini Notebook

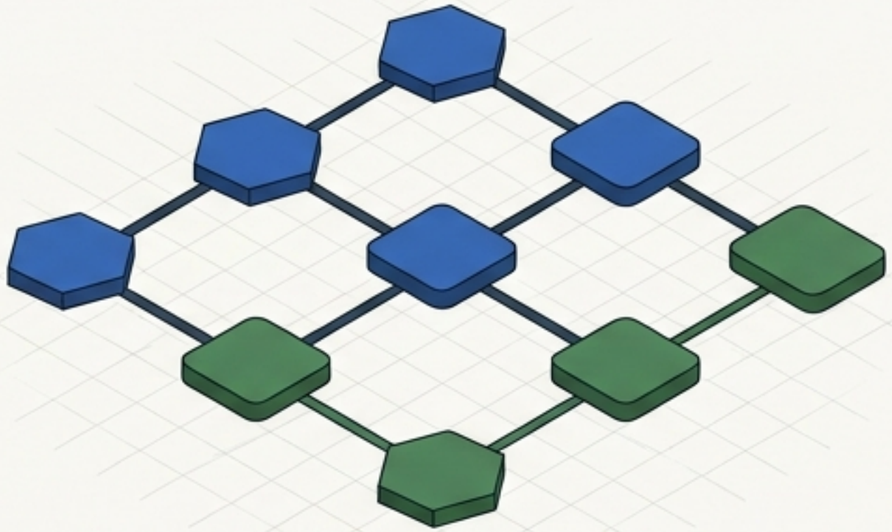
The end of siloed artificial intelligence

Standard Claude/Copilot



- **Knowledge:** Siloed to the individual developer's machine.
- **Learning:** Repeats the same traps for every developer.
- **Memory:** Auto-memory is hidden, unstructured, and private.
- **Governance:** Implicit team rules that must be manually prompted.

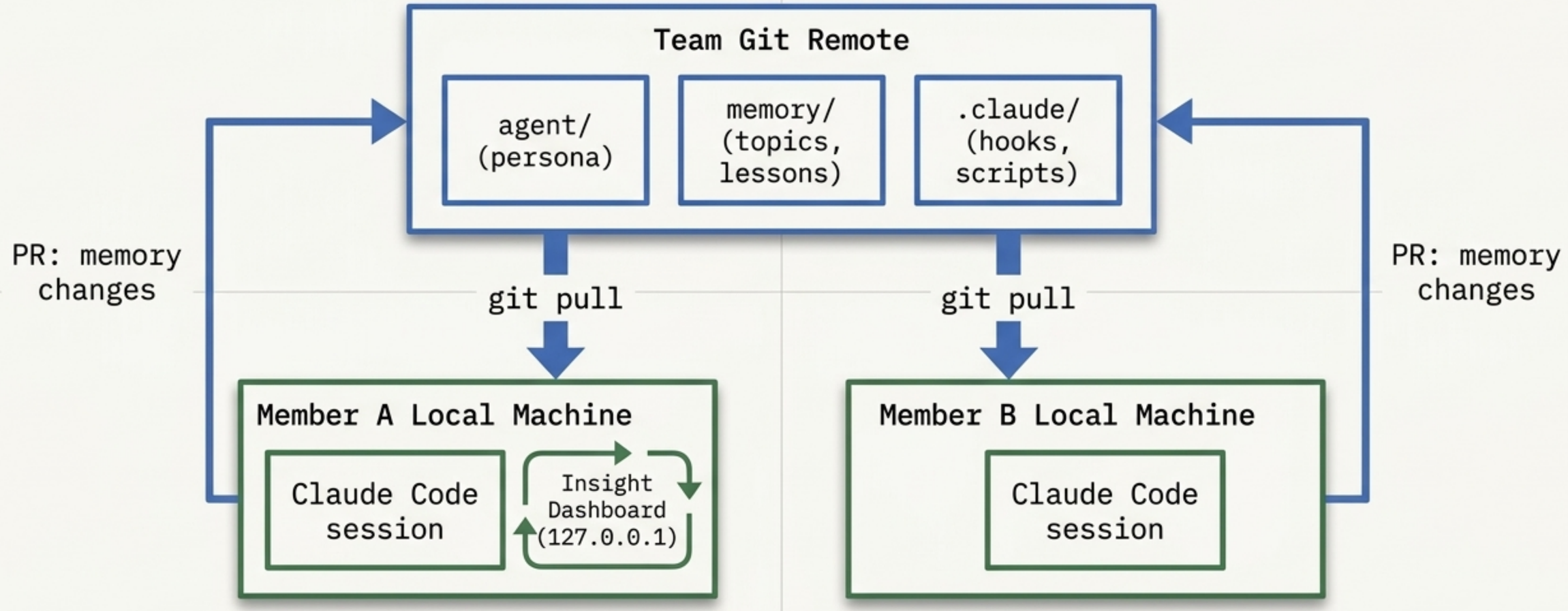
Agent-Team-Harness



- **Knowledge:** Shared team memory via a standard `Git remote`.
- **Learning:** One developer's learned lesson protects everyone.
- **Memory:** `PR-reviewed`, `structured markdown` files.
- **Governance:** Explicit team rules injected automatically at session start.

Gemini Notebook

The architecture requires zero new infrastructure.



The repo acts as the single source of truth for the persona, memory, and hook code. Everything runs locally on standard primitives: Linux, Python 3.10+, Git, and Bash. Nothing new to host.

Gemini Notebook

Every tool call and prompt flows through **deterministic gates.**

1. SessionStart

`load-persona.sh` injects the Tier 0 persona and trigger questions into context.

2. UserPromptSubmit

The prompt passes a Librarian Gate, fetching a relevant memory brief before the agent acts.

3. Guarded Work

Every tool call passes `PreToolUse` hooks (e.g., **git-guard** preventing destructive commands).

4. Turn Ends

The session pauses or loops back to the next prompt.

5. SessionEnd

The developer uses `/wrap` to enrich the daily log, extract lessons, and generate a memory-only PR.

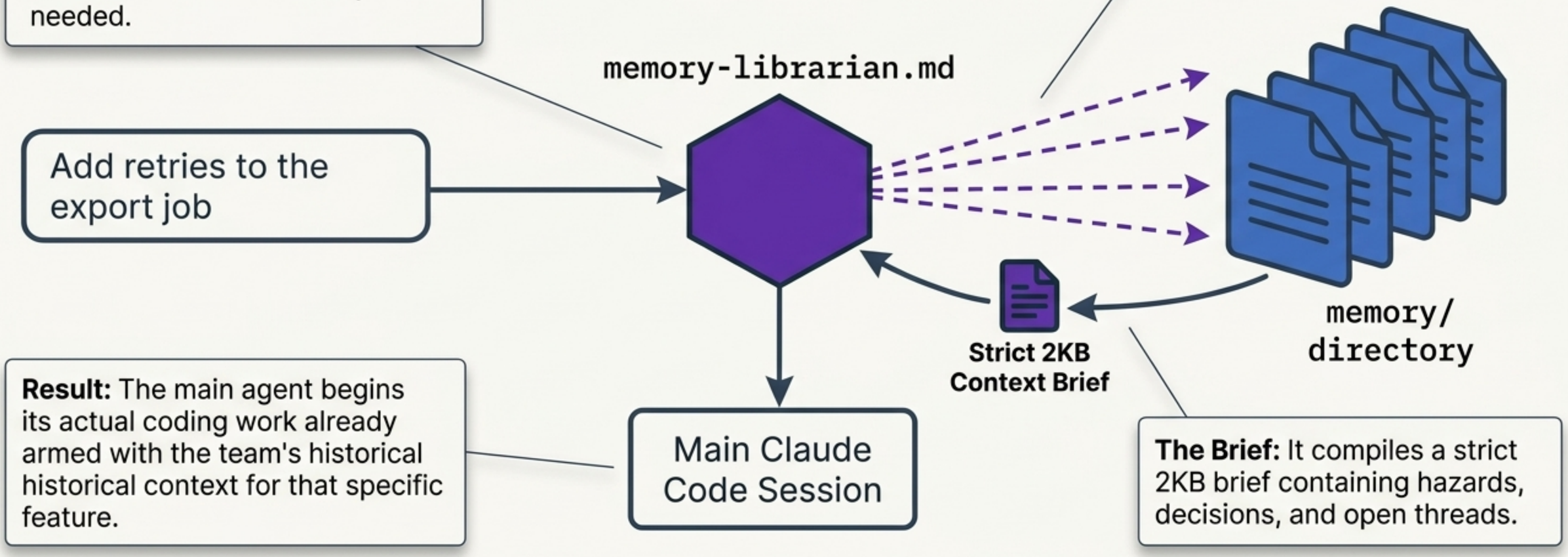
Gemini Notebook

The **memory-librarian injects context** before the agent begins work.

Anatomy of a Process

Intercept: The librarian gate temporarily halts the main prompt to evaluate if team memory is needed.

Search: The memory-librarian subagent independently searches the local cloned memory directory, reading only top hits.



Result: The main agent begins its actual coding work already armed with the team's historical historical context for that specific feature.

The Brief: It compiles a strict 2KB brief containing hazards, decisions, and open threads.

Gemini Notebook

A single developer's mistake instantly becomes a team-wide guardrail.

1. The Trap: Dev A hits a roadblock (e.g., a stale CI log).

2. The Extraction: Dev A runs `/wrap`. The agent writes a lesson file with a trigger question: *Did the step that WROTE this log run in this chain?*

3. The Review: Dev A opens a PR. The team merges the new lesson to `main`.

4. The Sync: Dev B runs `git pull`. The trigger question is automatically injected into Dev B's Tier 0 persona.

5. The Prevention: Dev B attempts a similar task. Dev B's agent automatically asks the trigger question out loud before falling into the same trap.



Gemini Notebook

Team memory is structured, human-readable plain text.

Topics (/topics)

Durable facts and decisions.

```
name: export-job-retries
type: decision
description: The export job retries 3x. Never retry 4xx.
```

Lessons (/lessons)

Trap prevention with explicit triggers.

```
name: check-the-log-writer
type: lesson
status: active
trigger: Did the step that WROTE this log run in this chain?
```

Logs (/log)

Daily chronological developer activity.

```
name: 2026-01-20.ada-lovelace
type: daily-log
author: ada-lovelace
```

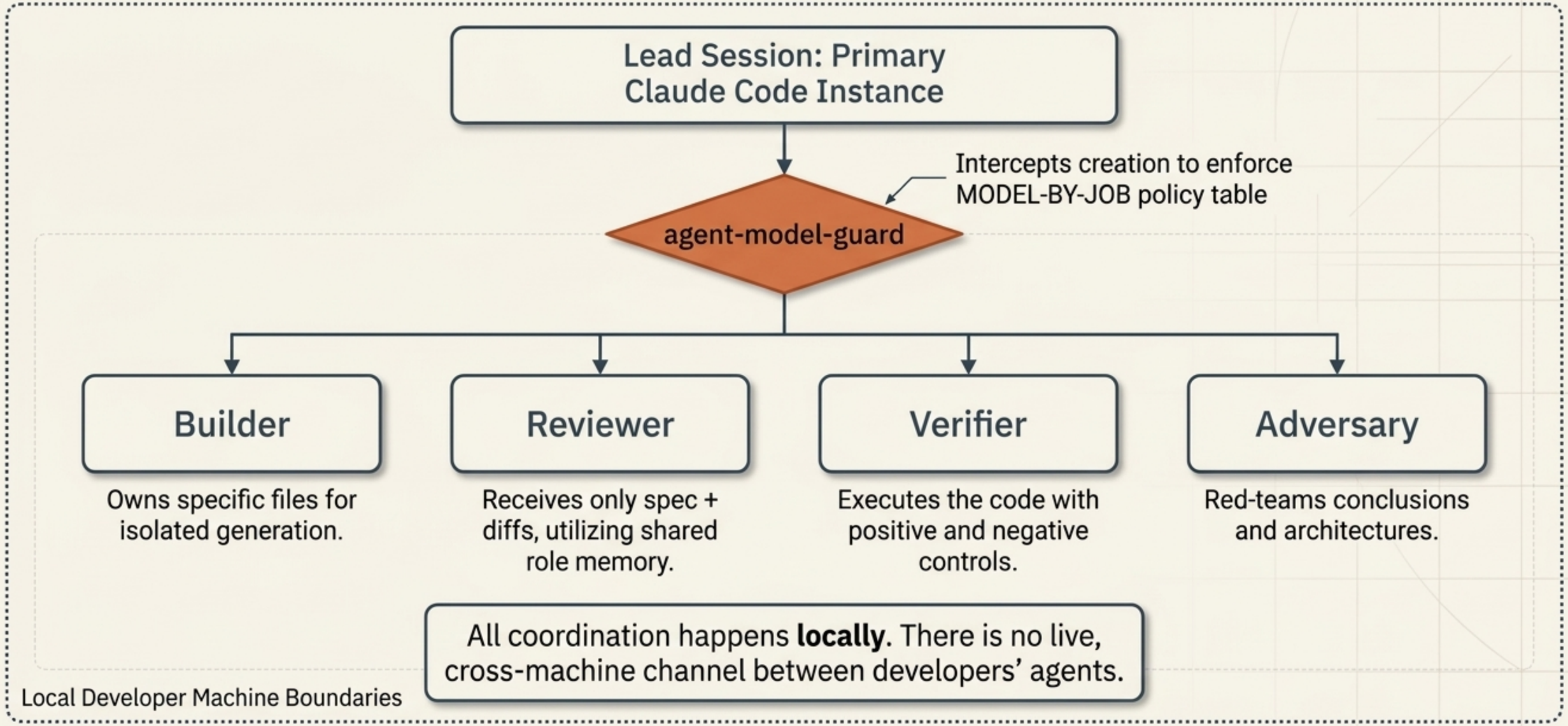
Generated Roles (/agents)

Lessons dynamically compiled into specialized subagents.

```
# Auto-compiled via agent_gen.py
# Pulls lessons tagged: review, verification, red-team
```

Gemini Notebook

Complete agent teams execute securely within a single local session.



Gemini Notebook

Strict trust boundaries isolate local work from shared logic.

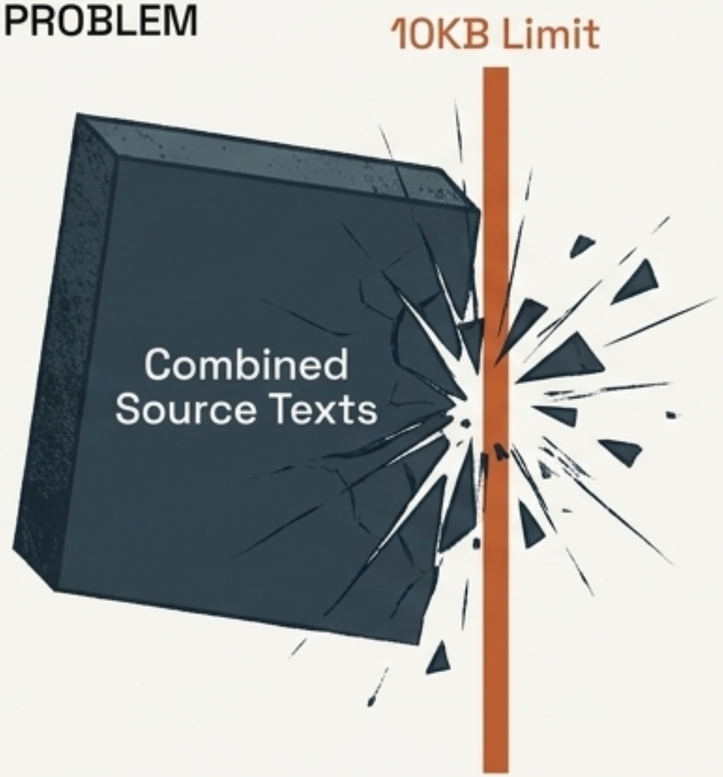


- **Transcript Privacy**: Claude Code auto-memory and session transcripts never leave the local `~/.claude` directory.
- **Code Integrity**: The `git-guard` hook explicitly blocks destructive commands (`stash`, `reset --hard`, `clean`) if the working tree is dirty.
- **Leak Prevention**: The `leak_scan.py` script scans against local patterns (e.g., client names, real emails) before any memory PR is generated.
- **Execution Control**: Hook code updates require explicit GitHub `CODEOWNERS` approval, treating prompt changes like production code.

Gemini Notebook

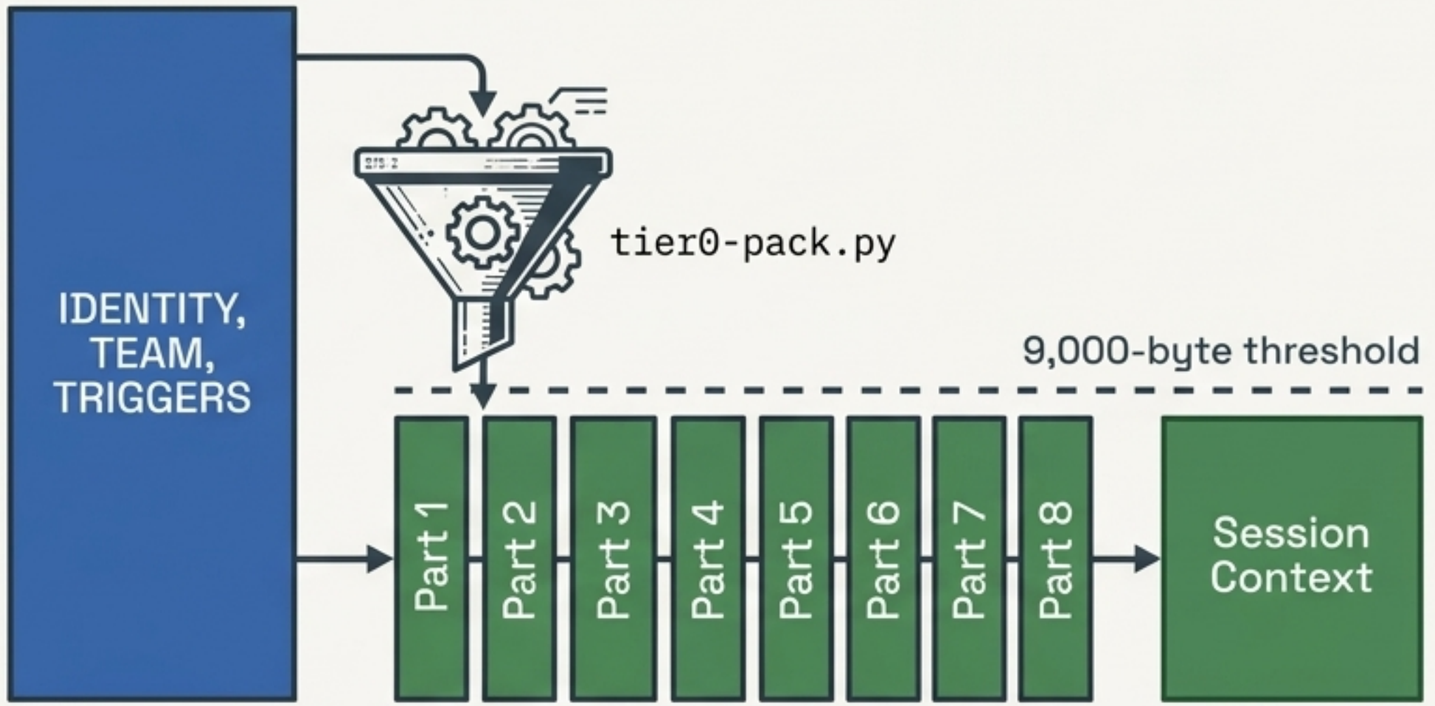
Bypassing context limits through deterministic chunking.

THE PROBLEM



Claude Code restricts hook output to 10,000 characters. If exceeded, the agent silently receives a useless file preview instead of its identity.

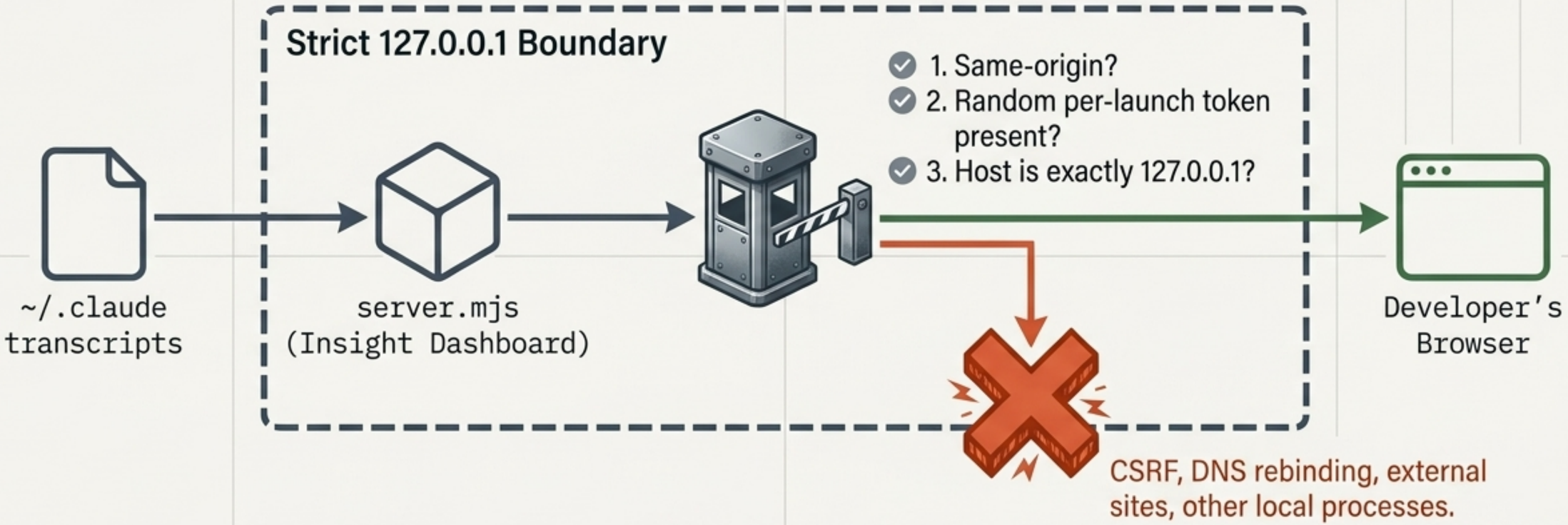
THE SOLUTION



The harness **deterministically slices the Tier 0 persona into 8 discrete parts of exactly 9,000 bytes.** Part 1 contains the manifest, allowing the agent to instantly recognize if a part failed to load.

Gemini Notebook

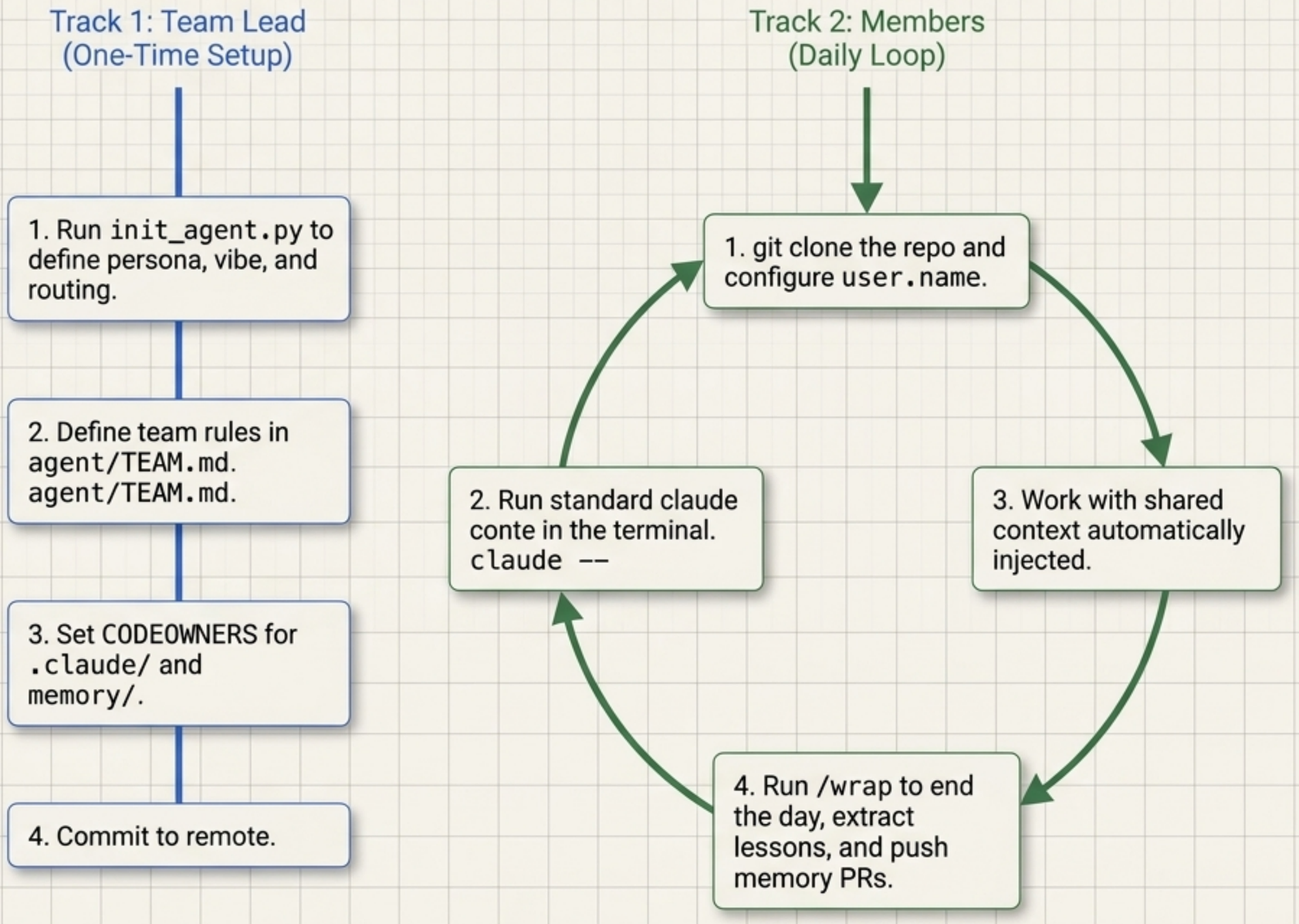
Local insight tracking blocked from cross-site exploitation.



Defense in Depth: Loopback binding is not enough on shared Linux hosts. Every /api/* request requires a random, per-launch token. The server enforces strict Host allow-lists, refuses cross-site requests, and never makes outbound calls.

Gemini Notebook

Integrating the harness into the daily engineering workflow.



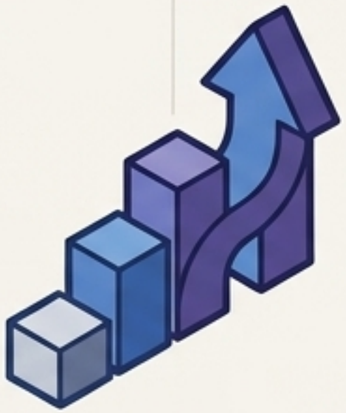
Gemini Notebook

Compound intelligence deployed on existing primitives.



Zero Infrastructure

Runs entirely on the tools your team already uses locally: standard `Git`, `Python`, and `Bash`. No new services to host, no external databases to manage.



Compound Intelligence

Your AI stops repeating the same mistakes. With every merged PR, the shared memory index grows, injecting preventative trigger questions directly into your team's workflow.



Secure by Design

Uncommitted code, local transcripts, and proprietary secrets never leave the developer's machine. Shared logic is governed by explicit explicit Git PR reviews and rigorous local hook guards.

Gemini Notebook